SEPTA SWARTHMORE STATION DISPLAY

Simple Case Build Guide

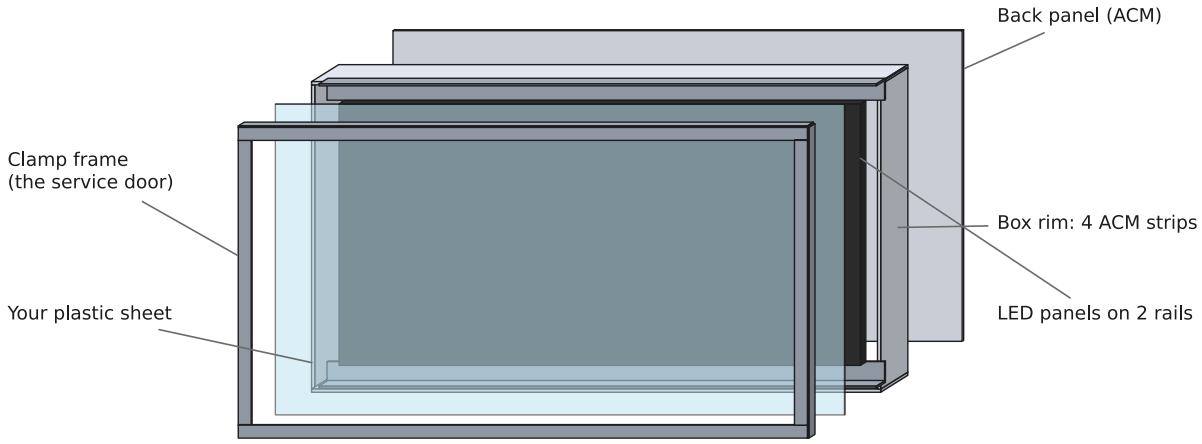
The quickest, cheapest way to get the board weatherproofed and on the wall: an aluminium-panel box you can build with a drill, a utility knife and a rivet tool, using the plastic sheet you already have as the window.

≈ $161
parts, not counting aluminium or a new window sheet

≈ 10 h
build time, about two afternoons

Easy
drill, utility knife, rivet tool, caulk gun

2 people
to hang it (about 50 lb)



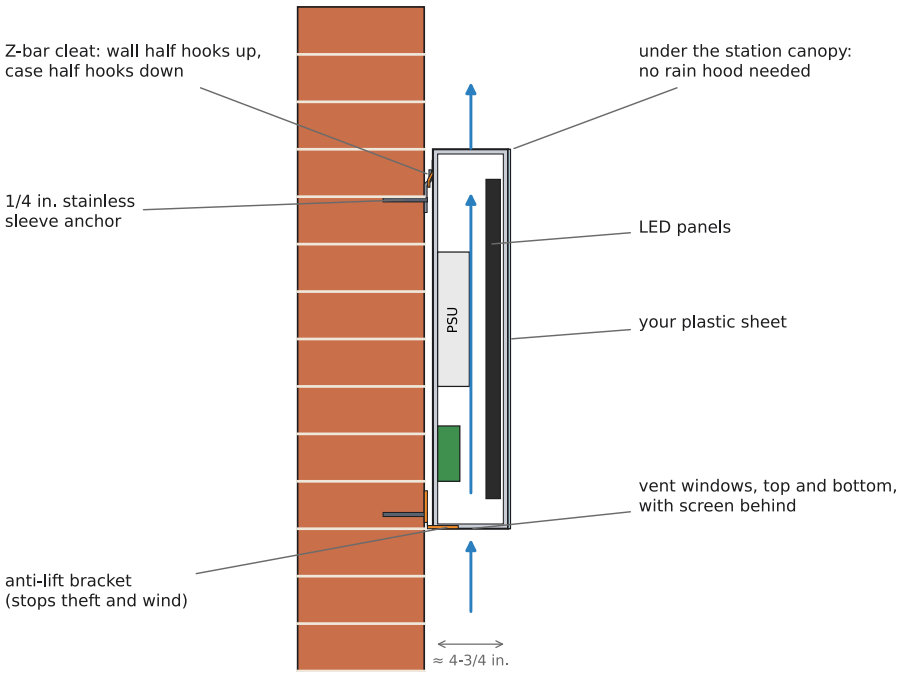
Before you drill the station

The building belongs to SEPTA, so mounting needs their written OK first, and they may want their own crew to do the drilling or the mains connection. The outlet is an electrician's job. Everything else in this guide is a normal workshop job.

How it works

The case is a shallow box made of **aluminium composite panel (ACM)**: two thin aluminium skins with a plastic core, the material most outdoor signs are made of. It cuts with a utility knife and holds rivets well.

- **Box:** four ACM strips form the rim, joined at the corners with short pieces of aluminium angle and pop rivets. An ACM back is riveted on and sealed, and stays on for good.
- **Window:** your existing plastic sheet sits on foam tape and is held by a **clamp frame** of aluminium angle, screwed into rubber **well nuts**. Removing those screws is how you open the case.
- **Inside:** the LED panels bolt to two rails across the top and bottom. The PSU and Pi screw to the back.
- **Air:** vent windows in the bottom and top let warm air rise straight through. The station canopy keeps rain off, so there is no hood.
- **Wall:** a 30 in. aluminium **Z-bar cleat** (one half on the wall, one on the case) carries the weight. Two small **anti-lift brackets** at the bottom stop anyone lifting it off.



What it leaves out, on purpose

No custom metalwork, no epoxy anchors, no powder coating, no fan, no rain hood (it hangs under the canopy). If it ever moves somewhere exposed, screw a strip of aluminium drip edge over the top vents on 1/2 in. spacers. If it runs hot in summer, or the brick turns out to be soft, the full plan (`docs/mounting-and-case-plan.pdf`) has the heavier-duty version of each part.

Before you buy: measure

Every size in this guide comes from your plastic sheet. The example numbers are for a **44 × 24 in.** sheet; fill in your own.

Measure	Example	Yours	Why
W: plastic sheet width	44 in.		Sets the case width
H: plastic sheet height	24 in.		Sets the case height
Panel array width (all 16 panels, edge to edge)	40.3 in.		At least 2 in. less than W
Panel array height	20.2 in.		At least 2 in. less than H
Panel array depth, LED face to back of the backplate brackets	about 1 in.		Sets where the rails go
PSU size	about 8.5 × 4.5 × 2 in.		Must fit behind the panels
Weight of the board as it is now (bathroom scale)	about 30 lb		The case adds about 20 lb

If the sheet is too small

The clamp frame covers 1 in. all round, so the sheet must be at least 2 in. bigger than the panel array each way, or the frame hides LEDs. If it isn't, order a new sheet cut to the array size plus 3 in. each way. It is in the shopping list but not counted in the total.

Cut sizes from your measurements

Part	Formula	Example (in.)	Yours
Back (ACM)	W × H	44 × 24	
Top and bottom strips (ACM) ×2	W × 4-1/2	44 × 4-1/2	
Side strips (ACM) ×2	(H – 1/4) × 4-1/2	23-3/4 × 4-1/2	
Back-edge angles, top and bottom ×2	W – 2	42	
Back-edge angles, sides ×2	H – 2-1/4	21-3/4	
Corner angles ×4	3	3	
Clamp frame, top and bottom ×2	W	44	
Clamp frame, sides ×2	H – 2	22	
Panel rails ×2 (thick angle)	W – 1/2	43-1/2	
Cleat backing bar (thick angle)	32	32	
Anti-lift brackets ×2 (thick angle)	2	2	

Shopping list

Prices from retailer listings in October 2026; check before ordering. The total counts parts only. Aluminium stock and a new window sheet are listed so nothing is forgotten, but are **not counted**.

Parts (counted)

Item	Qty	Where	≈ $
Z-bar cleat, 30 in. aluminium, rated 300 lb (Hangman HM-30D)	1	Home Depot	14
Stainless sleeve anchors, 1/4 in., about 2-1/4 in. long	6	Home Depot, stainless sleeve anchors	12
Aluminium pop rivets, 3/16 in., grip range, 50-pack	2	Arrow RLA3/16IP, Home Depot	16
Clear outdoor silicone, 10.1 oz	1	GE Silicone 2, Home Depot	11
Rubber well nuts, #10-32	14	Bolt Depot	26
Pin-in Torx security screws, #10-32 × 1/2 in., stainless, 25-pack	1	Albany County Fasteners	20
EPDM foam tape, 1/8 × 1/2 in. × 65 ft	1	Okayasu, Amazon	15
Cable gland, PG13.5 nylon (pack)	1	Baomain, Amazon	10
Nylon M2.5 standoff kit (for the Pi)	1	Amazon, search "M2.5 nylon standoff kit"	9
Stainless fasteners: M4 × 10 mm screws ×4 (PSU); #10-32 × 3/4 in. screws, nylon lock nuts and washers, about 15 each	1 set	Home Depot, Everbilt stainless drawers	12
Aluminium insect screen, a repair patch or offcut (about 2 sq ft)	1	Hardware store	6
Green 14 AWG wire (2 ft) and two ring terminals	1 set	Hardware store	6
Nylon washers, 1/4 in. (keep stainless off aluminium)	10	Hardware store	4
Parts total			**≈ 161**

Tools (borrow from the engineering shop if you can)

Tool	Used for	≈ $ if bought
Drill; 3/16, 3/8 and 1/2 in. bits; 1/4 in. masonry bit	rivets, well nuts, vent corners, wall anchors	6 (masonry bit)
Hand rivet tool (Arrow RT100M or similar)	every joint	22
Utility knife, fresh blades, metal straightedge, 2 clamps	cutting the ACM	8
Jigsaw with a metal-cutting blade	vent windows	7 (blades)
Hacksaw or mitre saw with a non-ferrous blade; file	cutting the angle, deburring	shop
T-25 pin-in Torx bit	security screws	6
Caulk gun, tape measure, square, level, marker	sealing, layout	5

If every tool has to be bought, add about $54, which brings the whole job to about $215.

Aluminium and window (listed, not counted)

Item	Qty	Where	≈ $
ACM, 3 mm, 24 × 48 in., white/black faces, 4-pack (uses 2)	1	Beyond LED Technology or Amazon	48
Aluminium angle, 1 × 1 × 1/16 in., 8 ft	4	Everbilt 800057, Home Depot	86
Aluminium angle, 1-1/2 × 1-1/2 × 1/8 in., 8 ft	1	Everbilt 802617, Home Depot	52
Aluminium angle, 1-1/2 × 1-1/2 × 1/8 in., 4 ft	1	Everbilt 6416, Home Depot	32
New window sheet, UV-protected polycarbonate (only if the old one is too small or damaged)	1	TAP Plastics, cut to size	100 to 200

The build

Fifteen steps. In every picture, **orange** is what that step adds. Work on a big table with a blanket on it so the ACM faces do not scratch.

○ parts ○ tools

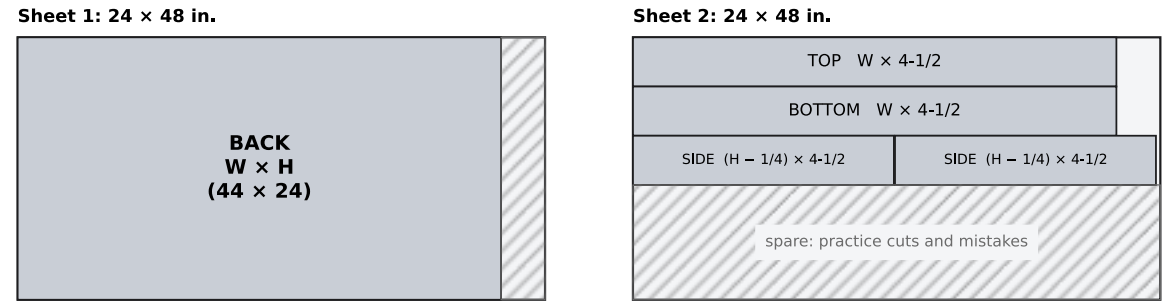
Two habits that save the whole build

Label every part with masking tape as you cut it. And dry-fit (hold it together with clamps) before you drill anything you cannot undo.

1 Mark the ACM

20 min

2 ACM sheets | tape measure | square | marker



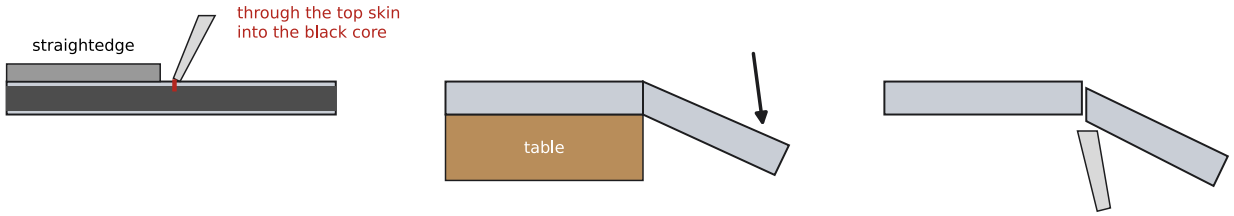
Example numbers for a 44 × 24 in. sheet. Strips are cut along the 48 in. length first, then to length.

1. Leave the protective film on, and mark on the film.
2. Sheet 1: mark the back, **W × H**, in one corner.
3. Sheet 2: mark four strips **4-1/2 in.** wide along the 48 in. length.
4. Pick which face goes outside (black looks best around a screen) and write "OUT" on every piece.

2 Cut the ACM: score and snap

45 min

marked ACM | utility knife, fresh blades | metal straightedge | 2 clamps | file



1 Clamp a straightedge on the line. Score 4 to 5 light passes, fresh blade.

2 Score line on the table edge, scored side up. Press the overhang down.

3 Run the knife along the bottom skin in the crease. File the edge smooth.

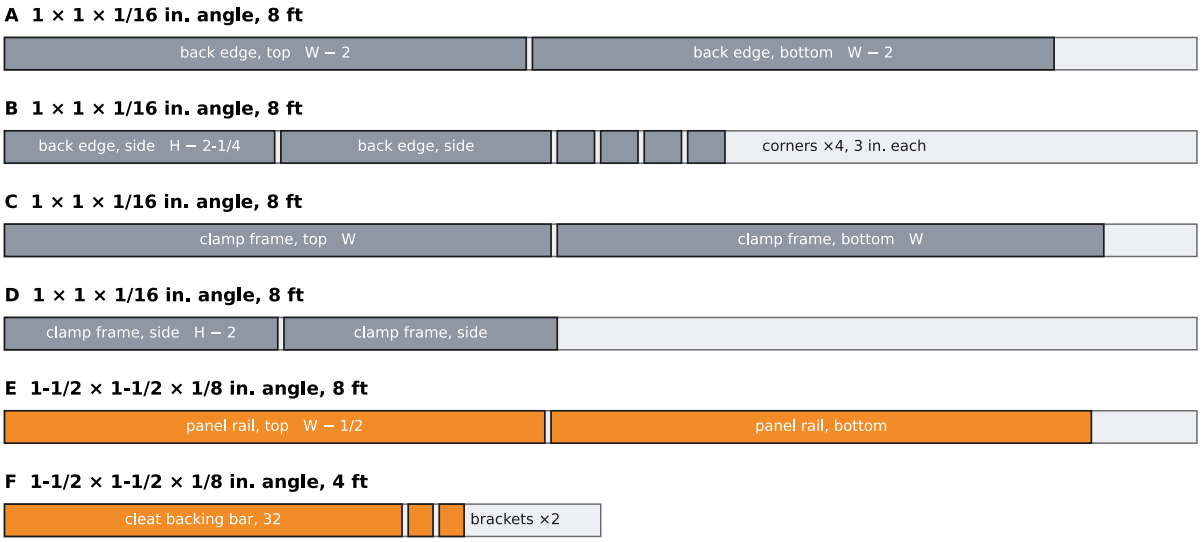
1. Clamp the straightedge on the line. Score **4 to 5 light passes**, until you are through the top skin into the black core.
2. Slide the sheet so the line sits on the table edge, scored side up. Press the overhang down firmly: it cracks along the line.
3. Cut the bottom skin in the crease with the knife, then file the edges smooth.
4. Cut the strips to length: top and bottom **W**, sides **H – 1/4 in.**

> **Tip**
> Practise once on the spare. Many light passes beat one heavy pass, which wanders off the line.

3 Cut the aluminium angle

45 min

4 thin + 2 thick angle sticks | hacksaw or mitre saw | file | masking tape



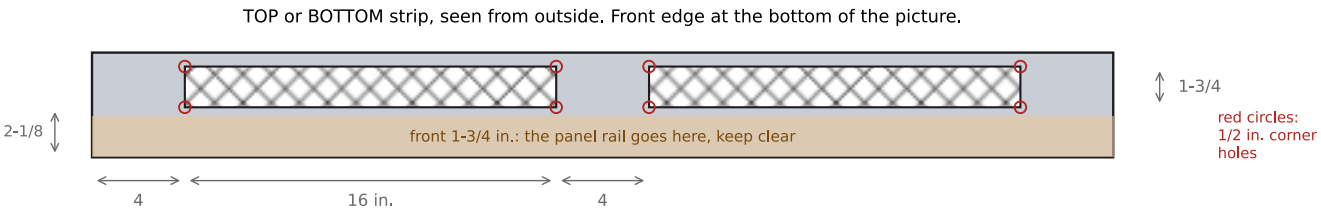
Grey: thin 1/16 in. angle. Orange: thick 1/8 in. angle (these carry the weight). Lengths in inches.

1. Cut the pieces shown. Every end is square: there are no mitres anywhere in this build.
2. File each cut end until it no longer catches a fingernail. Sharp aluminium cuts hands and wires.
3. Label each piece: "back top", "rail top", and so on.

4 Cut the vent windows

40 min

top and bottom strips | drill, 1/2 in. bit | jigsaw, metal blade | file



1. On the **top** and **bottom** strips, mark two 16 × 1-3/4 in. windows, 4 in. in from each end, starting 2-1/8 in. back from the front edge.
2. Drill a 1/2 in. hole inside each corner (red circles).
3. Join the holes with the jigsaw, supporting the strip on both sides of the cut so it does not chatter. File smooth.

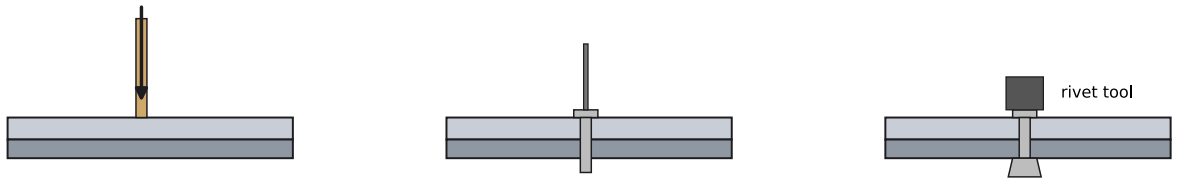
Why these sizes

The board gives off about 80 W of heat. Two windows per strip make about 56 sq in. of opening at top and bottom, enough for warm air to rise out on its own even with screen behind it.

5 Learn the pop rivet

10 min

scrap ACM and angle | 3 rivets | rivet tool | drill, 3/16 in. bit



1 Clamp the parts together. Drill 3/16 in. through both.

2 Push the rivet in from the side you can reach. Flange flat.

3 Squeeze the tool until the pin snaps. The tail swells behind and locks.

1. Clamp the two parts together and drill **3/16 in.** through both at once, so the holes line up.
2. Push a rivet in from the side you can reach, flange flat on the part.
3. Put the tool on the pin and squeeze until it snaps. Do three on scrap before the real thing.

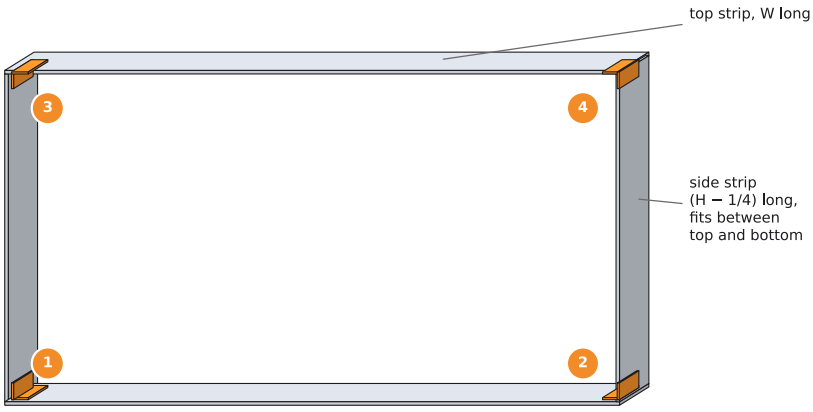
Spacing

Every 6 in. along long joints; 2 per leg on short pieces. If a hole misses, drill a fresh one beside it and rivet both.

6 Build the rim

45 min

4 ACM strips | 4 corner angles | 16 rivets | rivet tool | square | clamps

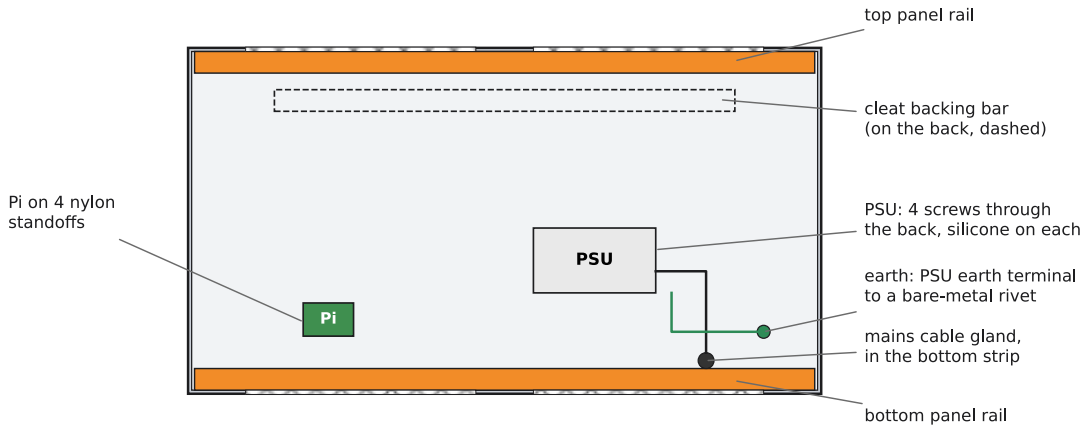


1. Stand the four strips on edge, "OUT" faces outward. The sides fit **between** the top and bottom strips.
2. Put a 3 in. corner angle inside each corner (1 to 4), centred in the depth, and clamp.
3. Check the corner is square, then drill and rivet: 2 rivets into each strip at each corner.
4. Measure both diagonals. If they differ by more than 1/8 in., push the long diagonal in and re-rivet that corner.

7 Rivet in the panel rails

40 min

2 thick rails | 12 rivets | the panel array | rivet tool | clamps

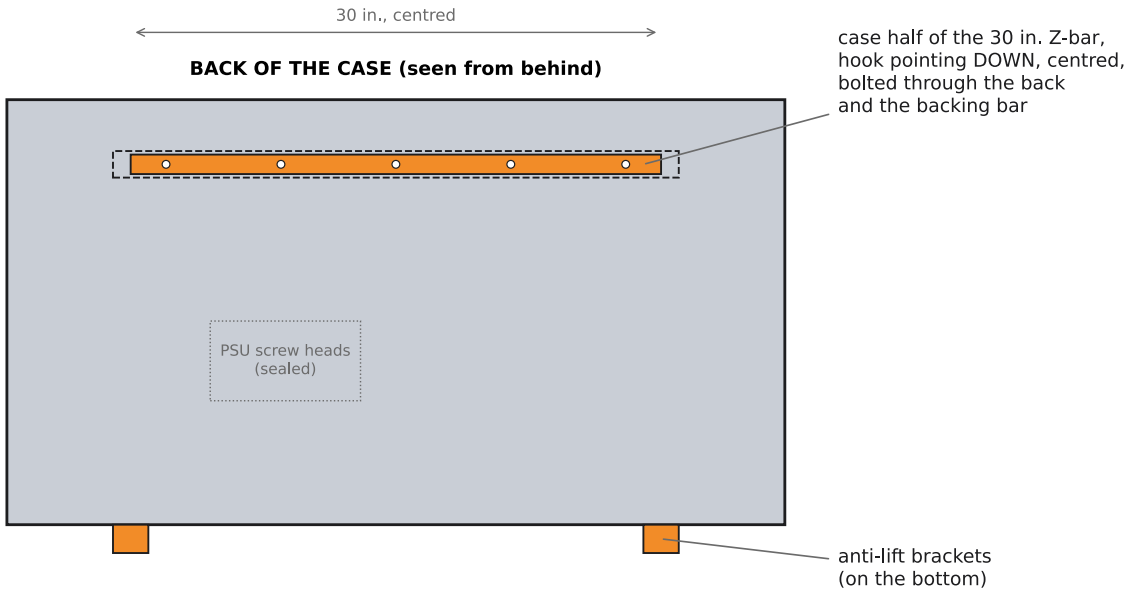


1. **Dry-fit first.** Lay the rim face-down, set the panel array inside, and find where its backplate brackets can bolt to a rail across the top and one across the bottom. Mark the rail depth so the LED faces end up **about 1/4 in. behind the front edge**.
2. Rivet each rail along the inside of the top or bottom strip: one leg flat on the strip, the other standing up to take the backplate bolts. Six rivets per rail.
3. Keep the rails in front of the vent windows, not over them.
4. Drill the rails for the backplate bolts now, while you can still reach.

8 Prepare the back panel

60 min

back panel | backing bar | Z-bar, case half | #10-32 screws, lock nuts | nylon washers | PSU | drill



1. **Cleat:** on the outside, centre the case half of the Z-bar near the top, hook pointing **down** (towards the floor once hanging). Put the 32 in. backing bar behind it on the inside. Drill through all three and fit 5 bolts: nylon washers under the heads, lock nuts inside.
2. **PSU:** hold the PSU where step 7's picture shows it, mark its 4 mounting holes, drill, and fix it with M4 screws from outside, a dab of silicone under each head.
3. **Pi:** drill 4 holes for the nylon standoffs where you can reach them from the front.

> **Hook direction**
> Hold the back up the way it will hang before you drill. The case half hooks **down**, the wall half hooks **up**. Getting it backwards is the classic mistake.

9 Screen the vents, fit the cable gland

30 min

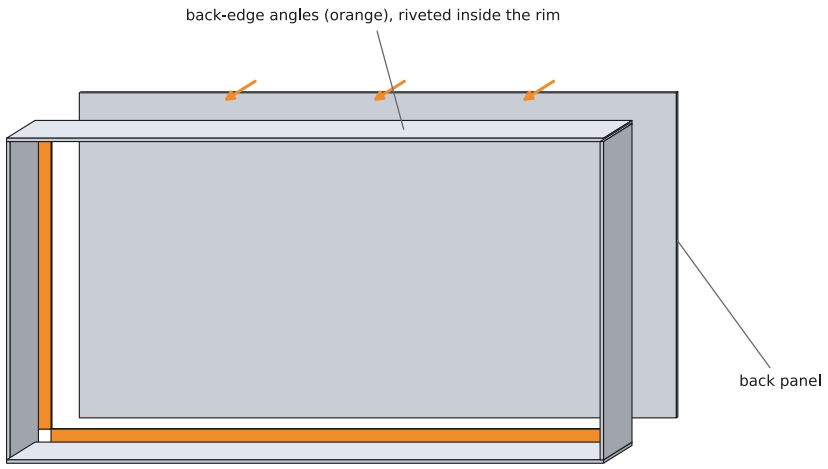
insect screen | silicone | cable gland | drill, 13/16 in. (or a step bit) | scissors

1. Cut four screen patches 1 in. bigger than each vent window all round.
2. Run silicone around each window **on the inside**, press the screen into it, and let it set.
3. Drill a 13/16 in. hole in the bottom strip below the PSU and fit the cable gland, nut on the inside.

10 Fit the back

60 min

rim | back panel | 4 back-edge angles | about 40 rivets | silicone | earth ring terminal | rivet tool



1. Rivet the four back-edge angles inside the rim, flush with its back edge: one leg on the strip, one leg facing back.
2. Run a bead of silicone along the back-facing legs.
3. Press the back panel on (cleat outside, hook down). Rivet through the back into the angles every 6 in. Wipe off the squeeze-out.
4. Put the earth ring terminal under one of these rivets, near the PSU, on bare metal: scrape the paint off at that spot first.

11 Electronics and wiring

60 min

Pi + bonnet | standoffs | earth wire | 16 AWG red/black | screwdrivers | wire strippers

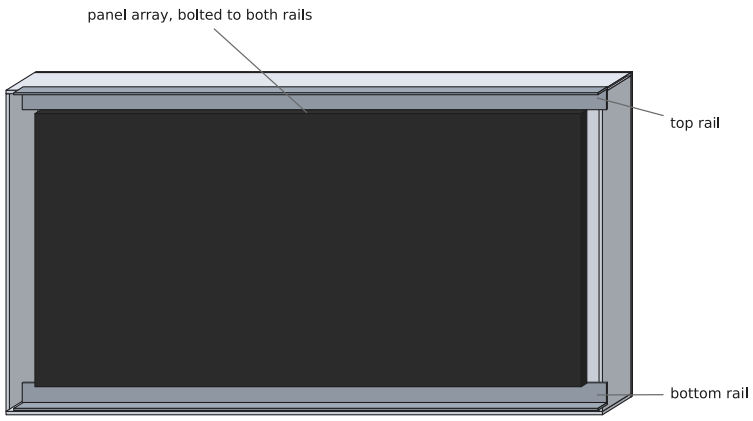
1. Fit the Pi on its standoffs.
2. With the cord **unplugged**, run it in through the gland with a drip loop outside, and tighten the gland.
3. Connect the green earth wire from the PSU's earth terminal to the ring terminal from step 10.
4. While you are in here, give the Pi its own 16 AWG pair straight from the PSU terminals: the under-voltage fix in `HARDWARE.md`.

> **Mains**
> Only the cord passes through the gland; the PSU's mains connections stay as they are. Anything on the 120 V side is an electrician's job.

12 Panels in, first light

45 min

panel array | backplate bolts | wrench

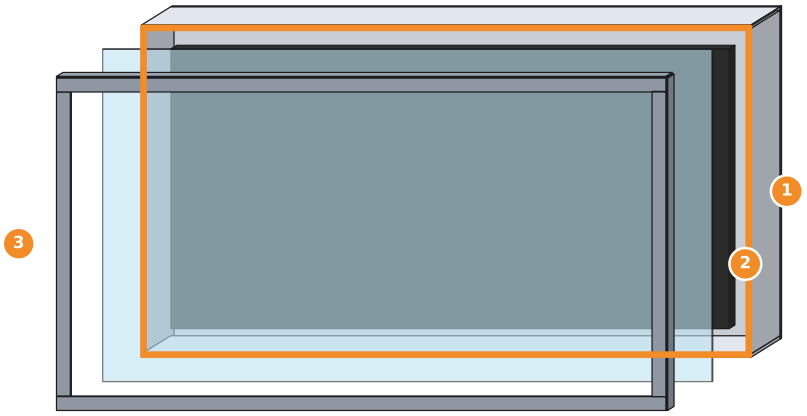


1. Lower the panel array in and bolt its backplates to both rails. Route ribbons and power so nothing rests on a sharp edge.
2. Plug in and run the board. Check every panel lights, the right way up, before you close anything.
3. Leave it on for 20 minutes and look for flicker on a white screen (the power test in `HARDWARE.md`).

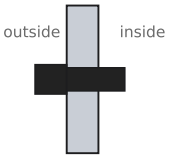
13 Window and clamp frame

60 min

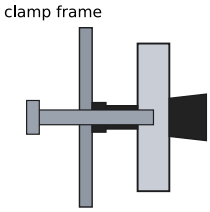
plastic sheet | foam tape | clamp frame (4 angles) | 12 well nuts | 12 security screws | drill, 3/16 + 3/8 in. | T-25 pin bit



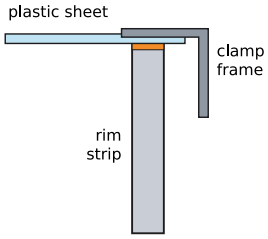
1 foam tape on the rim's front edge 2 your plastic sheet 3 clamp frame, screwed into well nuts



1 Push the well nut into a 3/8 in. hole from outside. Flange stays outside.



2 The screw passes through the clamp frame; the rubber bulges behind and grips. Watertight.



3 Cut-away: the sheet is squeezed between two strips of foam tape (orange).

1. Stick foam tape all round the rim's front edge (1). Clean the plastic sheet and peel any film.
2. Lay the sheet on the tape (2). Stick foam tape inside each clamp-frame piece's front leg.
3. Set the clamp frame on (3): top and bottom pieces full width, side pieces between them. Clamp it.
4. Drill 3/16 in. through the frame's side leg and the rim: 4 holes along the top, 4 along the bottom, 2 down each side.
5. Take the frame off. Open just the **rim** holes to 3/8 in. and push a well nut into each from outside. Refit the frame and drive the screws until the foam squeezes about halfway. **No tighter:** acrylic cracks if it is clamped hard.

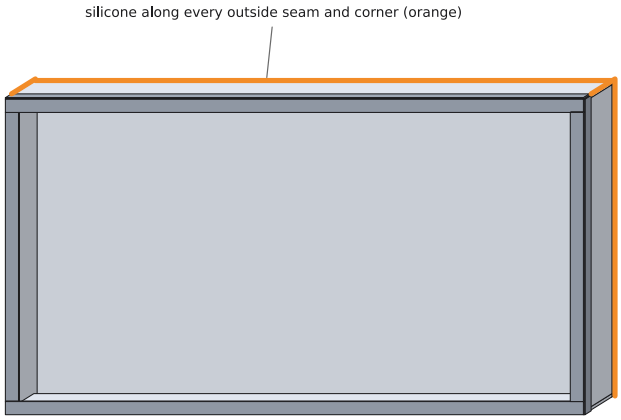
No holes in the plastic

Nothing goes through the sheet. It floats between two strips of foam, so it can grow and shrink with the seasons without cracking.

14 Seal it

30 min, then overnight

silicone | caulk gun | masking tape, rag



1. Mask both sides of each outside seam. Run a thin bead of silicone where the back meets the rim, and down each corner.
2. Smooth it with a wet finger, pull the tape, and let it cure overnight.
3. **Do not** silicone the clamp frame. Its foam tape seals it, and it has to come off for service.

15 Test before it goes up

3 h, mostly waiting

Test	How	Pass
Earth	Multimeter from the plug's earth pin to the back-edge angle and to the PSU case	Near 0 Ω (it beeps)
Rain	Garden hose on the shower setting, on the top, sides and window for 10 minutes. Then open the frame and look.	Dry inside
Heat	Closed up, normal board, 2 hours. Then `vcgencmd measure_temp`	Pi under 75 °C
Power	`vcgencmd get_throttled` after a white-screen test	`0x0`
Weight	Bathroom scale	Write it on the back

Hanging it

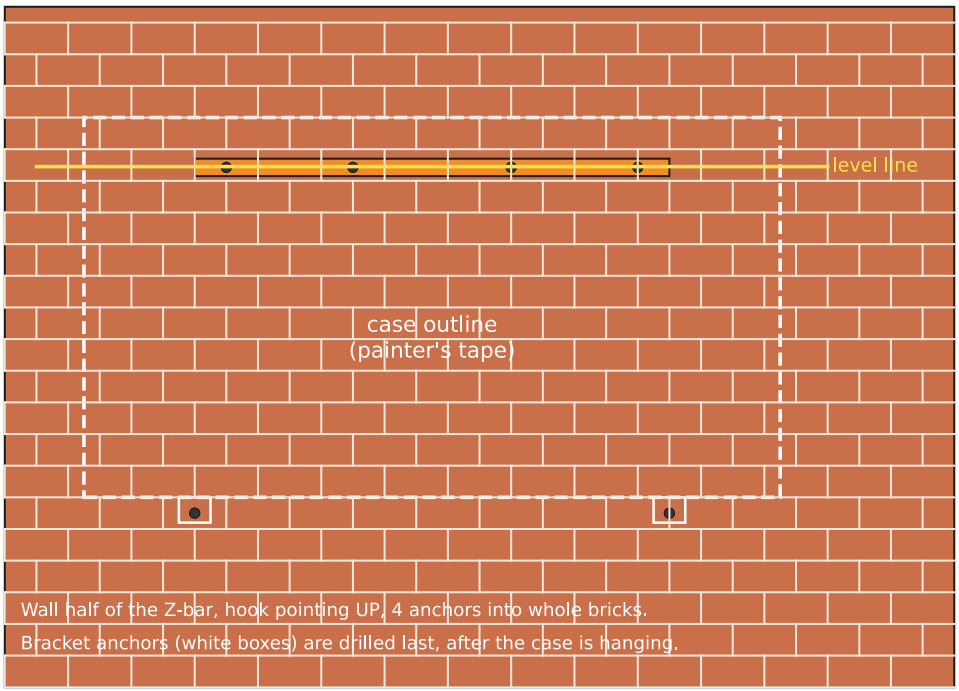
Before you drill

SEPTA's OK in writing; two people; two ladders or a platform. Drill one test hole low and out of sight first. If the dust comes out red and the hole stays crisp, the brick is solid and sleeve anchors are fine. If it is crumbly or breaks into a hollow, stop: use the epoxy method in the full plan instead.

A Mark the wall

20 min

wall half of the Z-bar | level | painter's tape | tape measure



1. Tape the case outline on the wall, bottom edge 7 to 8 ft above the platform.
2. On the case, measure from its top edge down to the cleat. Draw a level line that far below the top of the outline.
3. Hold the wall half on the line, hook pointing **up**, centred. Mark 4 holes, each in the middle of a whole brick (or in mortar joints, if SEPTA prefers).

B Anchors and the wall cleat

30 min

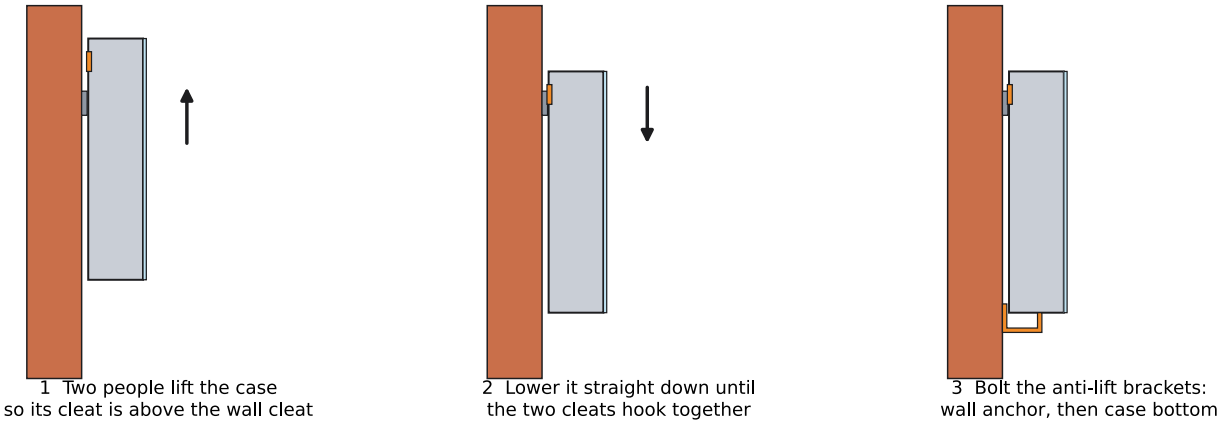
4 sleeve anchors | nylon washers | drill, 1/4 in. masonry bit | hammer | wrench

1. Start each hole with the drill on **rotary** (hammer off) so the brick face does not chip, and go 1/2 in. deeper than the anchor.
2. Blow the dust out. Tap each anchor in through the cleat (open the cleat's holes with a drill if the anchors don't pass).
3. Nylon washer under each nut; tighten until snug, then one more turn. Check the cleat is still level.

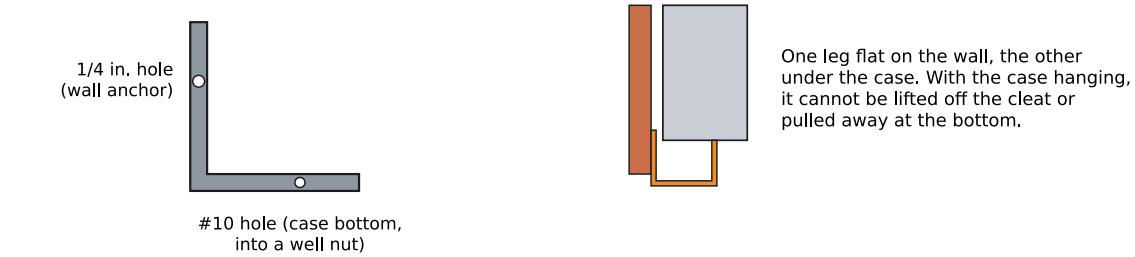
C Hang it and lock it

30 min

the case | 2 anti-lift brackets | 2 sleeve anchors | 2 well nuts, 2 security screws | 2 people



Anti-lift bracket: a 2 in. long piece of 1-1/2 in. angle (side view)



1. Two people lift the case until its cleat is just above the wall cleat, then lower it straight down. It drops about 1/2 in. as the hooks engage.
2. Hold each anti-lift bracket in place (one leg on the wall, one under the case), mark, then drill the wall for a sleeve anchor and the case bottom for a well nut.
3. Fit the anchors, then the security screws. Now it cannot be lifted off or pulled out at the bottom.
4. Plug into the GFCI outlet (fitted by the electrician) with a drip loop, and press TEST, then RESET.

Looking after it

When	Do this
You need to get inside	Undo the 12 clamp-frame screws (T-25 pin bit) and lift off the frame and sheet. Everything is reachable from the front once the panel array is unbolted.
Water inside after rain	Look for a missed silicone seam at the back corners, then for flattened foam at the window. Re-seal; replace foam that has taken a set.
It runs hot in summer	Check the vent screens are clear. If it is still hot, the full plan adds a small fan at the top vent.
Window scratched or cloudy	Mild soap and water only. No glass cleaner: ammonia ruins acrylic and polycarbonate. Replace with UV polycarbonate if it has yellowed.

When	Do this
Taking it down	Two people: undo the two anti-lift screws, lift up 1/2 in., pull away.

Every spring

- ☐ Clean the window and the vent screens
- ☐ Look over the silicone and the foam; re-seal anything cracked
- ☐ Check the anchors and anti-lift screws are tight
- ☐ Press TEST and RESET on the GFCI outlet

> **Source files**
>
> Built from `docs/simple-build/` in github.com/BenGreenberg07/septa-tracker: `guide.html` for the words, `figures.py` for the pictures, and `./build.sh` to make the PDF. The heavier-duty design is `docs/mounting-and-case-plan.pdf`.